



Preserving cultural landscapes: an integrated framework for the conservation and sustainable development of Mang Thit Pottery Village, Vietnam

Thinh Duy Do¹

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Abstract

Mang Thit Pottery Village (Vinh Long Province) is a river-based cultural landscape shaped by clustered linear ensembles of kiln-drying yard-house/workshop-wharf along the Co Chien River. Using qualitative spatial analysis, VES, semi-structured interviews, and document review, this study (i) identifies the village's operational spatial logic and (ii) translates it into an implementable conservation-planning agenda. Findings show that production, household life, and river logistics co-depend on riparian microclimates and short walking links to wharves; protecting kiln structures alone is therefore insufficient. We synthesise these results into the HLE framework (Heritage-Livelihood-Ecology), which treats heritage assets, social-spatial practices embedded in the landscape, and riverine ecology as one operational system. This framework underpins an integrated heritage-based planning model with four components: heritage continuity, experiential tourism, participatory development, and smart technology. Rather than converting Mang Thit into a display-oriented “heritage park,” the paper recommends recalibrating planning toward a living heritage corridor that keeps production in situ, maintains water access, and manages visitor loads by cluster. Priority actions include establishing 30–50 m riparian buffers with access easements, designating cluster-based conservation districts, and monitoring operational indicators. The proposed approach demonstrates how Mang Thit can move beyond architectural preservation toward sustainable cultural landscape conservation, aligning identity, livelihoods, and ecology over time.

Keywords Mang Thit Pottery Village · Cultural landscape conservation · Traditional craft heritage · Sustainable rural development · Integrated spatial planning

1 Introduction

1.1 Background

Traditional craft villages are a core element in the socio-cultural fabric of rural Vietnam. They embody local knowledge, artisanal skills, spiritual practices, and spatial organisation rooted in the identity of each community. Beyond their economic role, craft villages also serve as cultural centres of rural life, reflecting the enduring relationship between

people, nature, and heritage across generations [45, 62]. Under the pressures of globalisation, industrialisation, and urban expansion, many traditional craft villages in Vietnam and around the world face the risk of decline, especially in the transmission of skills, the continuity of spatial organisation, and the cohesion of community structures. Deacon (2003) identifies traditional craftsmanship as an essential form of intangible cultural heritage that must be safeguarded, as it carries values, identity, and cultural memory through time [11]. Deacon [12] further stresses that cultural heritage loses its vitality if it no longer exists within a living community context. Without real people and real places, conservation risks becoming little more than static museum display [12]. Internationally, craft villages, whether weaving in Peru, washi papermaking in Japan, or pottery in Morocco, face similar challenges such as economic transformation, generational discontinuity, and competition from

✉ Thinh Duy Do
thinhhdd@hcmute.edu.vn; tinh.do@fulbrightmail.org

¹ Faculty of Civil Engineering, Ho Chi Minh City University of Technology and Education, 1 Vo Van Ngan Street, Thu Duc, Ho Chi Minh City, Vietnam

industrial products. Studies show that without integrated policies linking heritage conservation, sustainable livelihoods, and appropriate urban planning, craft villages risk becoming hollow shells, stripped of the cultural soul that once bound together production techniques, rituals, belief systems, and everyday spaces [29]. The greatest danger is not only the loss of traditional skills but also the erosion of community identity itself, where production techniques, rituals, beliefs, and living spaces are interwoven into a unified whole. Therefore, safeguarding craft villages in the era of globalisation requires a comprehensive approach that moves beyond technical measures and instead promotes strategies that integrate cultural, economic, and environmental sustainability at the local level.

Mang Thit Pottery Village in Vinh Long Province, a renowned centre for red-clay ceramics in the Mekong Delta, flourished from the early twentieth century along the Co Chien River. Its iconic beehive kilns created a distinctive riverside landscape while structuring both livelihoods and community settlement. Yet shifts in market demand, the pace of urbanisation, and weak planning orientation now threaten the survival of this heritage landscape and the continuity of its cultural value.

In the international arena, various countries have advanced innovative conservation models. Japan's Living National Treasures system protects not only individual masters but also the cultural environment in which traditions flourish [25]. In South Korea, craft-based tourism has revitalised villages such as Icheon pottery village while strengthening national branding [28]. India's Crafts Cluster Initiative has linked heritage preservation with job creation for disadvantaged groups [30]. In contrast, research on Vietnamese craft village conservation remains limited. Most existing works focus on cultural history or tourism development but rarely integrate architectural, ecological, and livelihood perspectives in a systematic way [6, 24, 40, 60]. Even studies on unique sites such as Mang Thit largely remain at the level of general appeals rather than in-depth analysis [61]. This lack of foundational data hampers effective policy development and misses opportunities to integrate craft villages into broader local development strategies.

Preserving traditional craft villages, exemplified by the case of Mang Thit, therefore requires a holistic approach that goes beyond technical solutions. It calls for strategies that connect cultural heritage with sustainable economic, ecological, and community development. Filling existing academic gaps on spatial structures, landscape values, and craft ecosystems can provide a critical basis for policymaking and heritage planning at both local and national scales. This urgency is reinforced by the Vietnamese government and Vinh Long Province, which have articulated clear strategic visions for the Mang Thit ceramic cluster. The

Ministry of Construction's *National Tourism Development Plan, 2021–2030, Vision 2045* identifies Mang Thit as a potential hub for cultural and craft tourism, with an emphasis on conserving and developing its historic red-brick kilns [61]. Building on this, the Vinh Long People's Committee has issued a *Master Plan for the Mang Thit Ceramic Kiln Area to 2045*, aiming to transform the heritage site into a regional and national cultural destination with international reach [35].

For this reason, research on the conservation and development of Mang Thit pottery village is not only of theoretical importance but also of practical significance. It can directly support cultural and ecological spatial planning, contribute to sustainable development strategies that integrate heritage and service-based economies, and ultimately help maintain the cultural identity of the Mekong Delta in a rapidly changing world.

2 Research status & constructive literature review

2.1 Research status

Traditional craft villages are integrated spaces where intangible cultural heritage, artisanal knowledge, architectural landscapes, and community livelihood structures converge. They play a central role in rural life across many Asian countries such as Vietnam, China, Japan, and Nepal. In the context of rapid globalisation and urbanisation, traditional craft villages are at risk of decline due to disrupted skill transmission, changing living environments, and the absence of comprehensive conservation strategies [12, 29].

Research in China and Nepal illustrates the growing importance of integrated approaches that combine conservation with development. Chan et al. [9], in their study of the entrepreneurial ecosystem in ancient villages of southern Anhui, China, concluded that community participation, local identity, and heritage space provide a foundation for sustainable growth. However, spatial planning and landscape considerations remain underdeveloped [9]. By contrast, Neupane et al. [34] successfully implemented an integrated masterplan for Tansen, Nepal, which combined the conservation of the Dhaka weaving tradition with tourism development, heritage landscape design, and livelihood improvement. This study represents one of the few systematic frameworks connecting spatial, cultural, and economic planning [34]. Another approach emphasises ecological landscape planning, as proposed by Nguyen et al. [38] in Phung Xa village near Hanoi, where water systems and ecological structures were found to be essential for sustaining traditional village spaces [38].

In Vietnam, existing studies have highlighted the role of craft villages in community economies and cultural tourism. However, conservation research remains fragmented, often focusing either on cultural-historical aspects or tourism proposals, while lacking systematic integration of architectural space, landscape value, and livelihood structures. For example, Giang [16], in his study of the ancient village of Phuoc Tich, argued that linking tourism and handicraft value chains can stimulate local economic development [16]. Yet his analysis was primarily economic and did not explore the spatial or ecological dimensions of craft villages. Nguyen [37], in research on Bau Truc pottery village in Ninh Thuan, one of the oldest Cham craft settlements, stressed the importance of cultural adaptation under globalisation [37]. However, the study did not provide a spatially integrated planning framework for sustainable development. Other works have approached conservation through ecological perspectives, such as Schnitzer et al. [52] in Dong Thap, who proposed clean production models to balance economic growth and environmental protection [52]. Even so, the spatial-cultural dimension of villages as integrated heritage landscapes was not addressed. Mang Thit Pottery Village, a unique architectural and cultural heritage site of the Mekong Delta, remains understudied in terms of sustainable planning. Tuan [61] emphasises that the lack of baseline data and interdisciplinary approaches limits policy-making and results in missed opportunities to embed heritage into local socio-economic strategies [61].

Current Vietnamese guidance and policies on cultural landscape preservation remain fragmented when applied to pottery villages. Instruments tend to prioritize listing and tourism promotion, with weak operational links to spatial planning tools, ecological landscape management, and livelihood safeguards [59]. Empirical gaps include: (i) spatially explicit core, buffer, and transition zoning tailored to production, residential, and riverine interfaces; (ii) carrying-capacity and visitor-flow management for workshop and jetty nodes; (iii) benefit-sharing mechanisms to avoid elite capture and protect artisans' time; and (iv) clean production and circular resource guidelines for clay, water, fuel, and waste in pottery processes. As a result, current frameworks often fall short of linking space, ecology, culture, and livelihoods within a single, testable planning logic for craft settlements.

Positioning conservation within the SDGs clarifies both normative aims and measurable outcomes. The proposed framework explicitly links to SDG 11.4 (protect and safeguard cultural and natural heritage), SDG 8.9 (sustainable tourism and decent work for artisans), SDG 12 (responsible production and circular resource flows in pottery value chains), SDG 6 (water use, wastewater, and sediment management along riverine systems), and SDG 13 (climate

resilience for flood-prone deltaic landscapes). These linkages enable site-to-province indicator design and monitoring (e.g., share of heritage-aligned livelihoods; water reuse rates; low-emission firing practices; seasonality-risk diversification) that move beyond descriptive conservation toward adaptive, accountable management [37].

Findings from Mang Thit serve as a site-level testbed that can scale into province-wide heritage spatial planning. Specifically: (i) Spatial typologies and pattern mapping (kilns—workshops—waterway nodes—walkable connectors) can be formalized as GIS heritage layers for Vinh Long; (ii) A conflict—mitigation matrix (crowding, commodification, seasonality, benefit distribution) can be institutionalized as a planning checklist applied across craft clusters; (iii) Operational tools—heritage impact assessment, core—buffer zoning, timed access and visitor-flow modeling, and clean-production guidance—can be embedded in provincial regulations and investment screening; and (iv) SDG-aligned indicators can be integrated into the province's monitoring framework under ongoing planning instruments (2021–2030, Vision 2045), providing a clear pathway from site evidence to policy design [59].

Drawing from both international and domestic research, it becomes evident that conserving traditional craft villages cannot be limited to preserving artisanal skills or reviving production activities. What is required is a comprehensive approach that integrates spatial planning, ecological landscapes, cultural heritage, and community livelihoods. Nevertheless, most existing studies still treat these factors in isolation, focusing narrowly on economic or cultural aspects without proposing spatial strategies at regional or national scales. Conservation models often remain descriptive rather than analytical, and few studies address the restructuring of village spaces as dynamic heritage entities. In Vietnam specifically, no in-depth research has yet examined the interrelations between architectural space, landscape values, and livelihood ecosystems in distinctive villages such as Mang Thit, where historical and cultural heritage, unique architectural forms, and community livelihoods intersect. For this reason, interdisciplinary research on the conservation of Mang Thit Pottery Village holds both academic and practical significance. It can provide critical support for ongoing planning initiatives in Vinh Long Province, particularly within the framework of the *National Tourism Development Plan, 2021–2030, Vision 2045*. This study therefore seeks to construct an integrated theoretical and practical framework for conserving and developing Mang Thit, grounded in the connection between cultural heritage, architectural space, community livelihoods, and ecological landscape planning. The goal is to advance sustainable development for the Mekong Delta in the context of contemporary challenges.

2.1.1 Constructive literature review

Studies on cultural landscapes emphasize the interwoven relationships between natural environments, production histories, and the cultural value systems of local communities. According to UNESCO's framework, cultural landscapes are categorized into three types: clearly defined designed landscapes, organically evolved landscapes (relict or continuing), and associative cultural landscapes based on spiritual or artistic connections between humans and nature [48]. These classifications enable space to be interpreted as a living archive of knowledge, technical expertise, and collective memory [15, 50]. Such approaches provide a useful theoretical lens for traditional craft villages like Mang Thit, where residential, production, and aquatic ecological spaces form an integrated cultural–ecological system. Recent research extends this perspective by incorporating landscape governance, co-creation of identity, and multi-layered value assessments encompassing cultural, livelihood, and ecological dimensions, underscoring that conservation should be integrated with spatial, ecological, and social planning [18, 44, 49].

Literature on architectural morphology and rural-urban form highlights the value of spatial reading methods such as pattern language [2], typomorphology, and Space Syntax [4]. These tools support the identification of underlying structures such as lane, riverbank, and dock networks; the classification of house, workshop, and kiln typologies; and the measurement of spatial properties including connectivity, accessibility, and permeability. Spatial structure indicators such as integration, choice, and connectivity offer empirical means to test hypotheses about the movement of workers, tourists, and goods, thereby linking spatial configuration with livelihood efficiency and heritage experience [26, 42]. In the case of Mang Thit, pattern-based approaches can inform proposals for spatial configurations that serve conservation and development objectives, moving beyond mere documentation of heritage.

Contemporary scholarship on sustainable heritage conservation has shifted from static preservation to adaptive governance, participatory processes, and the integration of ecological, economic, and urban planning. Current frameworks emphasize planning based on historic urban landscapes (HUL), linking heritage with tourism and livelihood value chains while considering carrying capacities, and using tools such as Heritage Impact Assessments and benefit-sharing mechanisms to empower communities [1, 54]. In craft-village contexts, international research advocates experiential tourism models, symbolic and cultural product restructuring, and circular-economy approaches that include clean energy, water reuse, and sustainable materials to avoid over-commercialization and authenticity erosion [7, 43].

These orientations provide a basis for constructing conflict-solution matrices that address seasonal imbalance, interest fragmentation, and tourist overloading, all of which are relevant to Mang Thit's development.

Synthesizing these three literature streams reveals persistent gaps. Many studies treat spatial, ecological, and livelihood dimensions in isolation. Integrated frameworks that can simultaneously interpret cultural landscapes, design spatial configurations, and incorporate governance and livelihood mechanisms remain limited. Operational toolkits tailored to riverside ceramic craft villages, particularly those that employ actionable conflict-solution matrices, are scarce. Addressing these gaps, this study integrates cultural landscapes, spatial patterns, and sustainable conservation into a unified planning framework for Mang Thit; maps key spatial patterns, including kilns, docks, water and land routes, and housing or production clusters, using structural indicators to guide conservation and development priorities; and proposes six groups of solutions organized within a matrix of conflicts, indicators, tools, implementing organizations, and data sources to operationalize sustainable village-level heritage conservation in the Mekong Delta context.

2.2 Objective and significance of research

This study analyses and evaluates the current conditions of architectural space, ecological landscape, and traditional livelihood systems in Mang Thit Pottery Village. Its purpose is to identify the core cultural, spatial, and livelihood values that define the identity of this unique heritage entity in the Mekong Delta. A key focus is to clarify how cultural heritage, community livelihoods, and ecological landscapes can be effectively integrated into comprehensive planning. The research highlights the essential values that require preservation, including artisanal techniques, settlement structures, religious practices and rituals, as well as riverine ecosystems. Building on this foundation, the study proposes an integrated planning model for the conservation and development of Mang Thit, linking heritage preservation with livelihood improvement and sustainable tourism in harmony with environmental protection. Ultimately, the research aims to formulate strategic policy recommendations that support heritage spatial planning for Vinh Long Province and the Mekong Delta region.

3 Materials and methods

3.1 Research design

This study primarily employs qualitative, exploratory methods; any background quantitative figures, if referenced, are

used solely for descriptive context and are not employed for statistical inference or forecasting. The approach is designed to surface themes, keywords, and narratives concerning the current status and potential for conserving and developing the traditional pottery village of Mang Thit. Accordingly, the manuscript's core evidence is textual and thematic, derived from field observations and semi-structured interviews. The study does not aim to produce statistical generalizations or long-term projections within its scope. The methodology was implemented through the following steps.

3.1.1 Field observation protocol

Fieldwork was conducted in three rounds (3 days per round) in March and June 2025, covering both the dry and wet seasons to capture seasonal variation. Observations focused on six site types: (1) traditional kilns, (2) workshops, (3) riverside dwellings, (4) brick-drying yards, (5) river wharves/boat landings, and (6) support spaces (storage, cart paths, market edges).

Using the visual encounter survey (VES) method, both quantitative and qualitative data were collected [8]. Observers walked along designated areas at set times, conducting visual observations to capture detailed information on built environments and user interactions. At each site, the team recorded approximate dimensions, materials, functions, adjacencies, visibility and enclosure, proximity to water and paths, as well as observed user flows (craftspeople, residents, visitors). Daily observation windows were 06:00 to 10:00 and 15:00 to 18:00 to capture production rhythms and mobility peaks. Each node was revisited at least twice across different rounds. All field notes, drawings, and photographs were later coded and classified by theme to support the analysis of spatial structures, livelihood dynamics, and landscape transformation.

3.1.2 Semi-structured interviews

The study conducted 25 interviews across two strata: (A) master artisans and long-practicing potters (12 participants, median experience ≥ 20 years) and (B) residents/households engaged in pottery-related livelihoods (13 participants). The sampling strategy combined purposive sampling, to ensure coverage by kiln type, workshop scale, and river adjacency, with snowball referrals to reach hard-to-access artisans. Sample size was determined based on thematic saturation [17, 21], with no new codes emerging after approximately

the 22nd–23rd interview, which was adequate to capture role diversity (kiln owners, workers, traders, service providers, heritage volunteers) and a wide experience range (from 5 to over 40 years).

Eligibility required at least 3 years of involvement in pottery-related work or residence within ≤ 300 m of kiln/workshop clusters. For context on knowledge depth, the team recorded years of experience and roles. The interview guide covered three themes: (1) cultural memory and perceptions of pottery heritage, (2) production organization, livelihood dynamics, and challenges, and (3) expectations, proposals, and future prospects (see Appendix 1). Typical duration was 30–60 min, and most interviews were audio-recorded with consent.

Transcripts were analyzed thematically through open coding, theme development into pattern families aligned with space, ecology, and livelihoods, and peer debriefing/audit. Codes were triangulated with observation logs and spatial traces, and illustrative quotations are anonymized (Table 1).

4 Research area

Mang Thit Pottery Village lies along the Co Chien River in Vinh Long Province (Fig. 1), an area rich in red clay deposits and well served by waterways. Pottery production emerged in the late nineteenth and early twentieth centuries, initially meeting household needs and regional trade. The combination of locally available clay, river logistics, and traditional dome-shaped beehive kilns enabled firing temperatures above 900°C and helped Mang Thit become the Mekong Delta's largest red-pottery center [13]. Mang Thit ware is characterized by a reddish-brown, unglazed, rustic surface, formed by hand or simple molds; typical products include jars, vats, basins, and large vessels for storage and landscaping [13]. After a peak between the 1950s and 1970s, demand declined due to plastic and metal substitutes, urbanization, and policy gaps, and many kilns shut down by the early 2000s [47]. As of 2024, about 120 legacy kilns remain along the river, mostly inactive, but increasingly recognized as a distinctive industrial-heritage landscape incorporated into the province's plan for a Red Ceramic Industrial Heritage Park linked to cultural tourism, as set out in the General Planning Scheme to 2045 [35].

Case selection is grounded in four determinant parameters that make Mang Thit suitable for testing an integrated and conflict-aware framework. Spatial and architectural factors include kiln typologies, kiln-dwelling-yard interfaces, and cluster connectivity. Cultural factors involve craft lineage, community memory, and the festival and OCOP ecosystem. Ecological factors concern river adjacency, water

Table 1 Demographics of interview participants

Demographic characteristics	Household	Age	Years of experience
Semi-structured Interviewees	25 (5F,15 M)	34–65	5–40

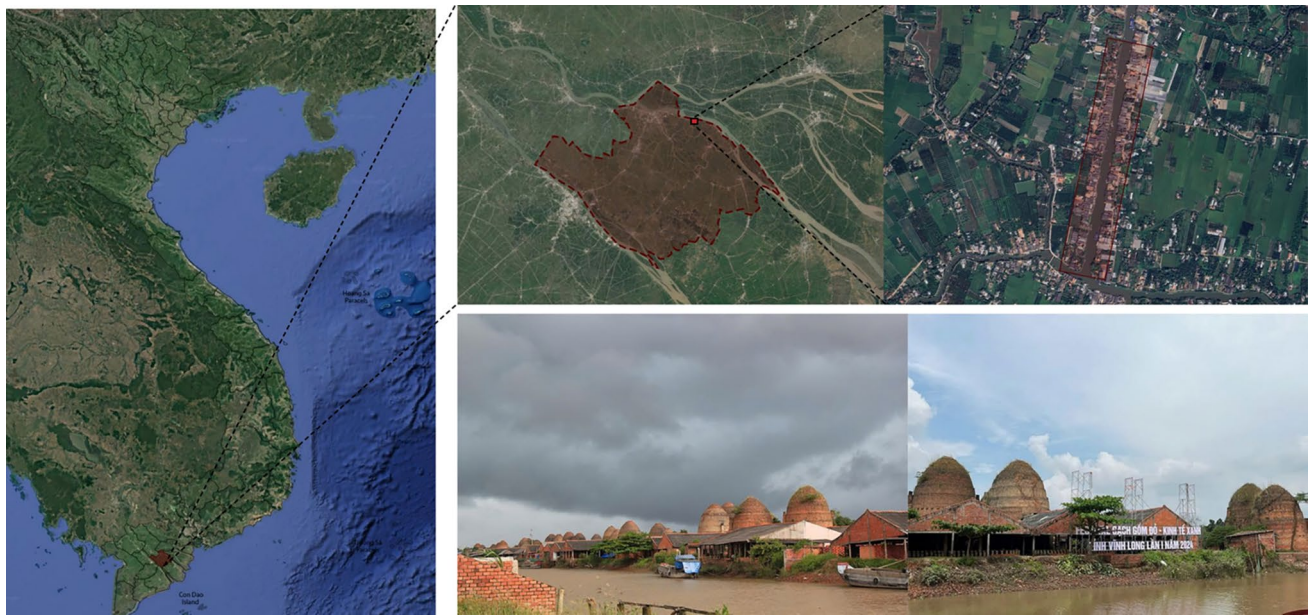


Fig. 1 Study area and existing conditions of pottery village

use and drainage, flood exposure, and clay sourcing. Livelihood factors include production diversification, tourism services, cooperative roles, and youth participation. Taken together, these parameters justify Mang Thit as a fit case for linking space, ecology, culture, and livelihoods within a single, testable planning logic.

5 Results

5.1 Analysis of the architectural space of Mang Thit Pottery Village

5.1.1 Overall structure and spatial clusters

The spatial structure of Mang Thit Pottery Village in Vinh Long developed primarily along the Co Chien River, where abundant red clay was easily accessible and river transport by boat served as the main means of movement. Unlike many pottery villages in northern Vietnam, which are typically organised in compact clusters, Mang Thit followed a linear waterway corridor with small residential groups connected to each hamlet and each kiln (Fig. 2). In this river-parallel form, kilns do not occur randomly but recur as short-interval ensembles that string together along the bank, forming a clustered-linear pattern rather than a single nucleated core.

Each kiln functioned as a production centre, usually associated with a single household or a group of families (Fig. 3). Around the kiln, different functional areas formed a shared living-production environment: drying yards

for pottery, family living spaces, working areas for shaping clay, and firing zones with distinctive beehive kilns. Interwoven into this environment were green spaces that provided shade, improved microclimate, and enriched the village landscape. These elements cohere into a repeatable spatial unit, namely “kiln, drying yard, house or workshop, and wharf,” which couples architecture, materiality (brick stacks, clay surfaces, shaded patches), and work rhythms (forming, drying, and firing).

The residential quarters were directly linked to both production activities and river wharfs, creating continuity between domestic life, labour, and trade. Together, these elements were enclosed within a system of tangible and intangible boundaries that defined the village’s spatial identity. Kilns were often positioned at relatively even intervals, stretching in a linear architectural pattern along the river-bank. Based on field mapping and high-resolution imagery, clusters were identified qualitatively by tracing the proximity of houses and workshops to their nearest kiln and the short walking connections from kilns to nearby wharves. This reading clarifies how circulation flows, including lanes and river access, stitch units into clusters along the corridor. The close integration of functional areas within this spatial structure produced a unique heritage landscape that reflects the deep relationship between pottery production, riverine ecology, and community life. This clustered-linear organization supports the interpretation of Mang Thit as an industrial cultural landscape, where repeating kiln-centered clusters scale household production up to a village-wide system through waterborne connectivity (see Figs. 2, 3).

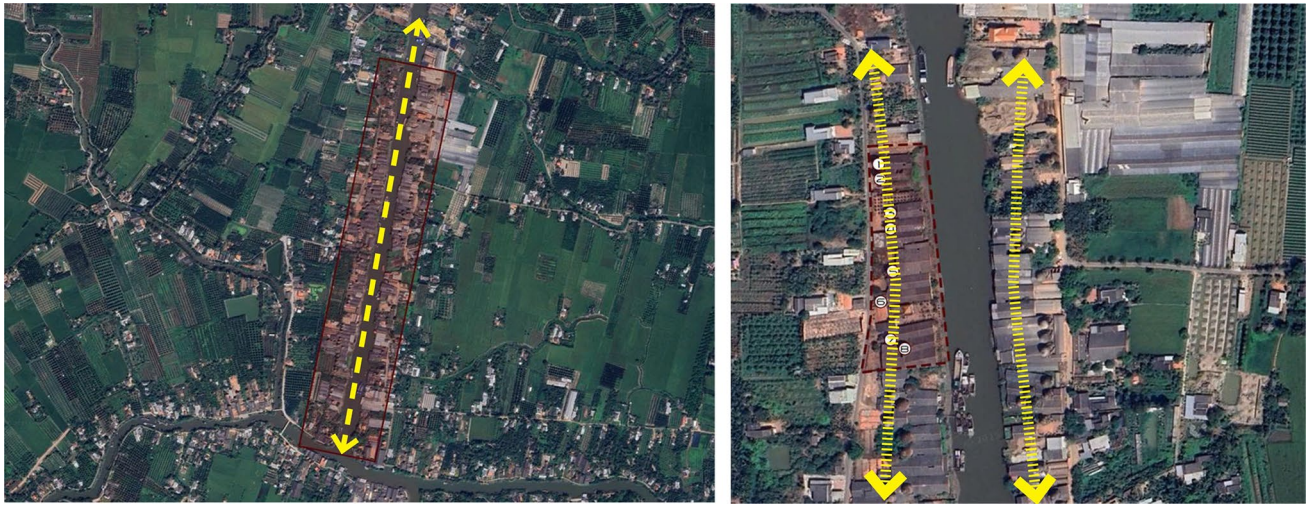


Fig. 2 Axial structure of Mang Thit Pottery Village arranged along both banks of the water body

5.1.2 Kilns, the core of the architectural space

The kilns and workshops of Mang Thit are arranged in several characteristic patterns. In the simplest type, a single kiln stands independently. A more compact configuration can be found in rows of three adjoining kilns that form a continuous block. Another variation places three kilns in a triangular arrangement, leaving an open courtyard at the centre. Some workshops are built in pairs, standing side by side and creating a semi-enclosed space. In other cases, a workshop is directly connected to a drying shed with either a pitched or vaulted roof. The most integrated form places a kiln at the centre of the workshop itself (see Fig. 4).

The most distinctive feature of Mang Thit's production architecture is the system of dome-shaped kilns, which possess both technical and morphological value. These kilns are typically semi-spherical in form, standing 8–11 m high, with walls 40–60 cm thick, built entirely by hand with raw bricks bonded with earthen mortar. Each kiln has a main entrance for loading products, evenly distributed vent holes, and a system of fire chambers and heating flues beneath. The base is compacted with soil or laid with rough concrete, with channels for hot air and smoke outlets beneath, ensuring stable circulation during the firing process (see Fig. 5).

The hemispherical or bell-shaped dome forms a self-supporting structure, allowing the kiln to withstand high pressure without the need for internal columns. At the same time, the design optimises the movement of hot air, making it possible to fire pottery and bricks at temperatures between 800 and 1000 °C. Each kiln can fire tens of thousands of items in a single batch, with the fire maintained continuously for 7–10 days.

Beyond their practical role, the kilns have become defining landscape symbols of the region. The rows of red domes

along the river create the striking image of an “industrial heritage field.” Yet today, more than sixty percent of the old kilns are no longer in use, with many deteriorating or dismantled for clay extraction, posing a serious challenge for conservation efforts.

5.1.3 Connection with the aquatic ecosystem

Mang Thit Pottery Village is not only a place that has preserved traditional ceramic techniques for more than a century but also a unique living and production space closely tied to the aquatic ecosystem at the heart of the Co Chien River. The river serves as more than a geographical feature; it functions as the central ecological infrastructure that determines the spatial organisation, production methods, and livelihoods of the community. In this sense, the Co Chien River can be understood simultaneously as natural heritage, cultural heritage, ecological infrastructure, and a locus of cultural practices that together constitute a broader cultural landscape.

The Co Chien River acts as a vital transportation axis. Boats carry red clay downstream from the upper regions, deliver firewood to the kilns, and transport ceramic products throughout the Mekong Delta and even to Cambodia and Thailand. Each kiln is located next to a river wharf, which provides facilities for loading and unloading goods, collecting water for washing clay, soaking soil, and cooling fired ceramics. During the dry season, water is also used to moisten drying yards, maintaining the necessary humidity for unfired products (Fig. 6). As natural heritage, the river's hydro-morphology supplies sedimentary resources and regulates the microclimate, enabling clay processing and drying; as cultural heritage, the persistent alignment of wharves, kiln-facing yards, and riverside dwellings materializes a long-standing boat-based craft organization.

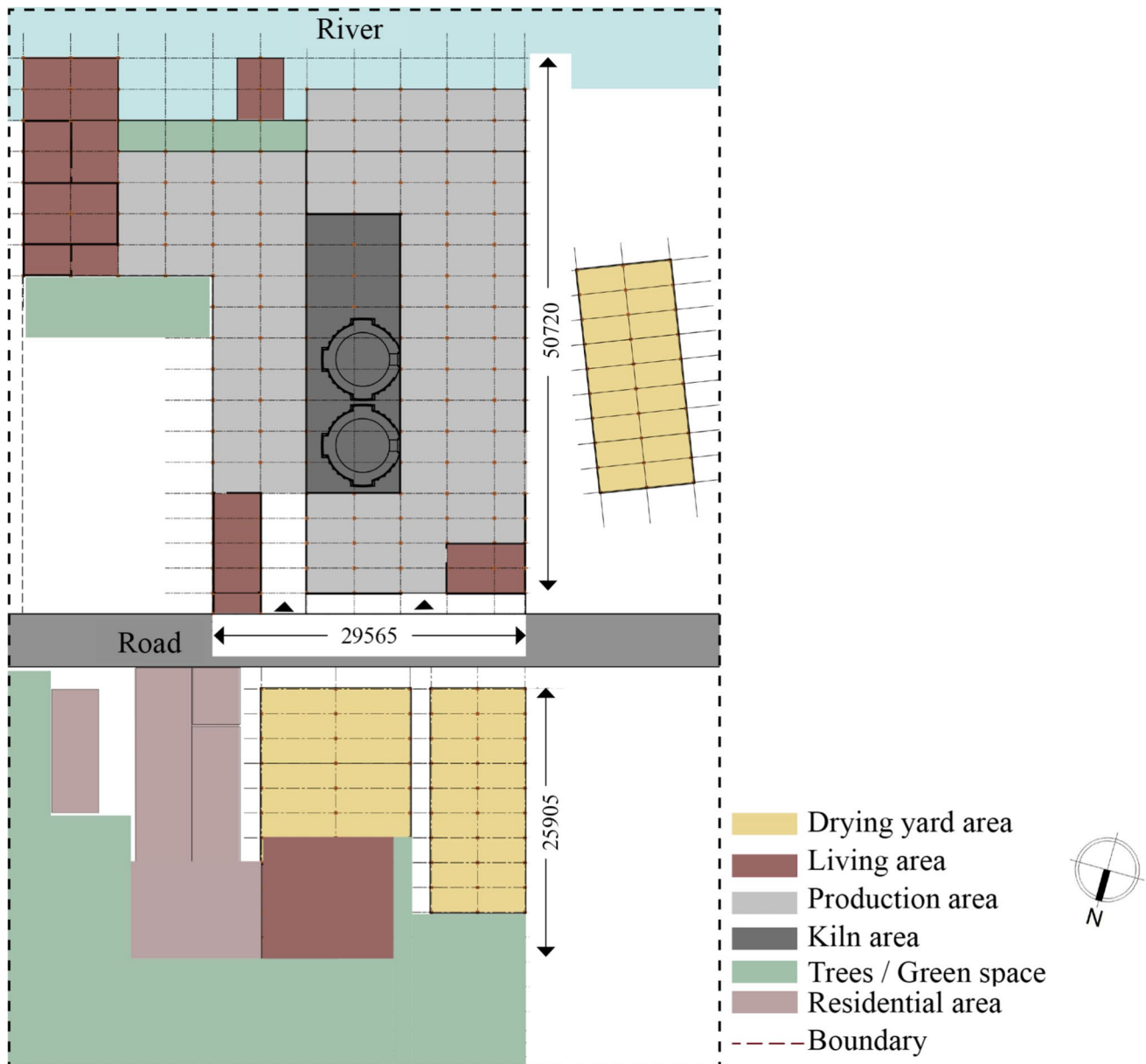


Fig. 3 Typical spatial layout of a pottery-producing household in Mang Thit Village

The wharf is directly linked to a characteristic production sequence that includes river landing, drying yard, kiln, and dwelling. Drying yards are used not only for displaying clay products and storing firewood but also as outdoor social spaces where children play, elders converse, and potters work while taking short breaks. The houses next to the kilns often serve as live-work spaces, reflecting a distinctive riverside settlement pattern in which several generations live together and pass on skills within the same household. Ecologically, riparian vegetation and open, permeable surfaces around wharves and yards help reduce heat stress, buffer wind, and stabilize moisture conditions that are critical for forming and drying; culturally, daily work rhythms, skill

transmission within extended households, and gatherings on the yards are synchronized with boat schedules and seasonal water levels.

These elements together form a village system that is interwoven with nature, making full use of hydrological conditions without degrading the landscape. They illustrate a form of harmony between people and the environment. In recent decades, however, urbanisation, encroachment on riverbanks for new housing or facilities, and the decline in demand for traditional ceramics have disrupted this balance. Many kilns have been abandoned, wharfs have been filled in, and the once resilient relationship between ecology, production, and community has been weakened. From a



cultural landscape perspective, these pressures fragment the functional linkages among kiln, yard, dwelling, and wharf, thereby diminishing both tangible values, such as riverside architectural patterns and landing facilities, and intangible values, such as boat-based craft practices and intergenerational learning.

for sustainable development based on local identity and ecological regeneration. In the current context of climate change and livelihood crises in rural riverine areas, such an approach represents both cultural preservation and a pathway for survival. Accordingly, management and design should prioritise the protection of riparian corridors and traditional landings, maintain water access for craft operations, and support community-led practices that align kiln operations, yard use, and household livelihoods with seasonal



Fig. 6 Community, production, and circulation along the Co Chien River, showing the coupled sequence of landing, drying yard, kiln, and dwelling, and the role of riparian vegetation and water access in supporting work rhythms and social practices

Fig. 7 Residential and living spaces of villagers in connection with production areas



hydrological cycles, thereby reinforcing Mang Thit's identity as an industrial cultural landscape.

5.1.4 Housing architecture and living spaces

The housing architecture in Mang Thit Pottery Village is an integral part of the production-living system characteristic of traditional craft communities. It represents a live-work model that needs to be preserved together with the pottery kilns. At present, however, this spatial continuity has been disrupted by unplanned residential buildings that encroach on riverbanks and fragment the production-living sequence.

Within the overall structure of the village, houses are not only places of residence but also play a central role in sustaining production spaces. Traditional dwellings of artisans and potters are typically modest, built with wood and covered by yin-yang tiles or corrugated sheets, and located close to the kilns (see Fig. 7). Their layout often consists of a main hall for living and working, opening onto a front yard used for drying products, storing fuel, or repairing pottery. This semi-open space serves multiple functions, accommodating both production and daily activities such as cooking, conversations, and the passing down of skills.

A defining feature of these houses is multigenerational cohabitation within one compound, which creates an interwoven pattern of living and working. As families expand, houses are often extended horizontally in a chain-like manner, with long eaves, connected corridors, and shared courtyards that foster intimacy and exchange among generations.

This live-work spatial model exemplifies the adaptability of southern Vietnamese craft villages to natural conditions and artisanal needs (see Fig. 7). Yet urbanisation and changes in land use are steadily eroding it. New constructions lacking coordination have broken the continuity of the traditional sequence of house, yard, kiln, and riverbank. Many riverfronts have been filled, and numerous kilns have been dismantled or abandoned, weakening the relationship between dwelling, production, and the river.

Safeguarding residential spaces in connection with the kilns is therefore essential to preserve the cultural identity of Mang Thit. Conservation should extend beyond physical structures to include the characteristic community model of living and working together. As a socio-spatial system of historical, cultural, and occupational value, it embodies the heritage of a riverside craft village in southern Vietnam and

offers important lessons for sustainable development in the future.

5.2 Community concerns and local perspectives

For residents and artisans in Mang Thit, pottery and brick-making is more than an economic activity; it is an inseparable part of both personal and collective identity. The traditional kiln clusters along Kinh Thay Cai, Hoa My, and Nhon Phu are recognised as symbols of space, livelihood, and heritage that have been closely tied to rivers and settlement practices for over a century. Kinh Thay Cai, a 12-km canal originally dug by local people, is remembered not only as an irrigation project but also as the backbone of village development, enabling the transport of raw materials and finished products. Before the year 2000, the village flourished, with hundreds of kilns operating continuously, producing up to 50 million bricks and pottery items annually and generating revenues of around 700 billion VND. Since 2005, however, the craft has declined due to high costs, industrial competition, scarcity of clay, and environmental restrictions. Today, only about 111 kilns remain in limited operation.

In this context, local concerns clearly reveal both challenges and expectations for the future (Table 2). Older generations fear the loss of traditional skills and call for stronger support to revive the craft as a living heritage. Younger workers, by contrast, show less interest due to unstable incomes, leading many households to shift to other occupations. At the same time, there is a strong demand for transforming production from low-value bricks to higher-value ceramic products such as decorative pottery and souvenirs. Residents also express pride in the cultural and historical significance of the craft and wish to see old kilns and workshops preserved as heritage landmarks. Community-based

tourism has emerged as another promising direction, with aspirations to develop guided tours and pottery workshops that combine cultural preservation with new livelihood opportunities. Yet concerns remain about the living environment, especially the health impacts of smoke from traditional kilns, highlighting the need for improved technical solutions that allow both heritage conservation and community well-being.

Despite these difficulties, local creativity and adaptation continue to be evident. Some households have introduced technical innovations, adopting mechanical equipment and circular rice-husk fuel systems that reduce labour and firing time while improving productivity and lowering environmental impact. These examples show the resilience and recovery potential of the craft in a changing context. Alongside technological adaptation, cultural and heritage tourism is gaining momentum. Many visitors express fascination when exploring historic kiln clusters, listening to stories of pottery making, and learning about generations who lived in close relationship with land, water, and fire. The riverside landscape of brick kilns, combined with hands-on craft activities and local culture, has opened new pathways for community-based and experiential tourism as a driver of the local economy.

6 Discussion

6.1 Heritage park or new-generation craft village? a choice requiring comprehensive consideration

The concept of a “heritage park” has become increasingly common in urban planning policies linked to cultural conservation in Vietnam. This model is conceived as an open space that combines the preservation of tangible and intangible heritage with tourism, education, and community recreation. According to Linh et al. [31], heritage parks represent a form of urban cultural reconstruction, where heritage is exhibited, interpreted, and reconnected with modern experiences [31]. In the planning framework for Vinh Long Pottery Village, the proposed transformation from a craft village into a heritage park has been promoted as a means of boosting tourism, revitalizing living spaces, and enhancing economic value. Indeed, the *Master Plan for the Tourism Area of Mang Thit Brick and Pottery Kilns, Mang Thit District, Vinh Long Province to 2045*, based on the orientation of the Vinh Long Provincial Plan, highlights particular concern for the heritage core with its distinctive historical, cultural, and landscape values. The spatial scheme divides the area into nine subzones that combine conservation, services, tourism, rural-urban residential areas, ecological landscapes, and agriculture, all linked through an interregional network

Table 2 Key concerns of Mang Thit pottery village residents

Key Concern	Details
Preservation of the traditional craft	Elderly residents wish to maintain the craft and fear its disappearance. Many expect government support to help revive the pottery village
Income and employment	Younger generations show little interest due to unstable income. Some households have shifted to other occupations
Transformation of production models	Demand for learning new techniques that combine tradition and modernity (e.g., decorative ceramics, handicrafts, souvenirs)
Cultural values and heritage	Residents express pride in the village tradition and hope for recognition and preservation of old kilns and workshops as heritage sites
Community-based tourism	Desire to develop pottery village tourism to generate additional income (organising tours, pottery workshops, etc.)
Living environment	Concerns about environmental pollution from traditional kiln smoke, requiring technical solutions

[35, 61]. This reflects an integrated vision that seeks to balance conservation, development, livelihoods, and tourism.

However, such an ambitious approach also raises critical concerns. The introduction of multiple mixed-use developments, urban residential clusters, and resort areas could result in excessive commercialization, weakening the authenticity of the heritage landscape. Moreover, the plan at present focuses primarily on functional zoning and does not clearly articulate a strategy for conserving brick kilns, pottery-making traditions, or traditional live-work models. The role of residents and artisans in the planning and management process is also insufficiently addressed, even though their participation is crucial for sustaining a “living heritage” rather than reducing the site to a form of “heritage on display.” These gaps indicate a need to recalibrate the heritage park concept toward a living system that retains production, residential uses, and river access within protected clusters. Proposed measures include designation of kiln conservation districts; adaptive reuse guidelines for inactive kilns; protection of riverside settlement morphology and landing facilities; and mandatory design codes for live-work house-workshop typologies.

Even during the current period of decline, the community remains deeply attached to its craft traditions, and the village space cannot simply be transformed into a museum-like park. This view is consistent with research by Anh and Ly (2024) in Hoi An, which cautioned that if local communities, especially artisans, are excluded from the process as heritage custodians, the very values intended for preservation will gradually be lost [59]. It is crucial to recognise that heritage is not only made up of artefacts or buildings but also encompasses ways of life, craft skills, and social practices, none of which can be “displayed” in the absence of living practitioners [14].

Transforming a craft village into a display-oriented park risks disrupting the traditional livelihood chain and displacing artisans from their role as active producers. Comparative studies of craft village conservation models in Cao Bang and Thailand by Tâm [56] demonstrate that when communities are excluded from heritage creation, both livelihoods and the continuity of intangible heritage are endangered [56]. Similarly, Binh [6] warned that conservation models not grounded in traditional production often result in superficial “cultural shells” that fail to safeguard core values [5]. In the case of Vinh Long Pottery Village, where pottery is embedded in inherited skills, labour practices, and ecological ties to the river, separating the craft from its cultural landscape would dismantle the entire socio-ecological-economic structure of the community. Accordingly, the livelihoods layer should include apprenticeship hubs and maker spaces; community-managed wharves; microfinance for

small-batch, high-value products; and equitable sharing of tourism benefits with artisan households.

Another critical but often overlooked factor in planning is the deep interconnection between cultural life and the local ecological environment. Anthropological studies in the Mekong Delta have shown that residents of craft villages do not simply make a living from the craft; they also inhabit the cultural space of the craft, where the river not only provides resources but also shapes rhythms of life, festivals, beliefs, and community structures [20]. Converting the craft village into a park risks breaking this relationship if the living, working, and cultural practices connected to nature are not sustained. Therefore, any restructuring design must take into account not only architectural concerns but also cultural, social, and ecological dimensions. For implementation, adopt 30–50 m riparian buffers with native vegetation, water-access easements for craft operations, low-impact mobility along the corridor, and visitor carrying capacity thresholds to protect both community life and river ecology.

6.2 Implications for integrated spatial planning of craft villages: conservation, development, and experiential approaches

Mang Thit Village in Vinh Long Province is widely recognized for its traditional red-brick kilns and longstanding ceramic craftsmanship. As urbanization expands and tourism interest grows, the village faces both challenges and opportunities. A thoughtful planning strategy can ensure that Mang Thit remains not only a heritage site but also a living, productive, and economically viable community. Building on findings from Sects. 3.1.1 and 3.1.3, which identify the clustered linear “kiln-drying yard-house/workshop-wharf” ensembles and their dependence on the Co Chien riparian system, this paper advances an integrated planning model that brings together four components: heritage continuity, experiential tourism, participatory development, and smart technology (Fig. 8). Guiding this approach is the HLE framework (Heritage–Landscape–Ecology), which interprets Mang Thit as a sustainable cultural landscape in which Heritage (kiln clusters, riverside morphology), Landscape (the walk-yard-kiln-wharf sequence, community practices, in situ livelihoods), and Ecology (the riparian corridor and microclimate) operate as a unified system. The integrated heritage-based planning model concretizes the HLE lens into the four components noted above, translating the patterns identified in Sects. 3.1.1 and 3.1.3 into implementable rules and tools (see Fig. 9; Table 3).

This model explicitly reframes conservation from a focus on structures to a sustainable cultural landscape, in which architectural fabric, household livelihoods, and riverine

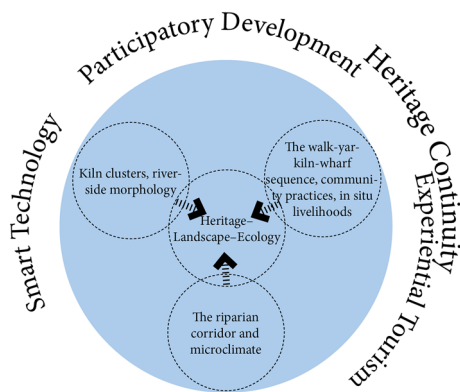


Fig. 8 Integrated heritage-based planning

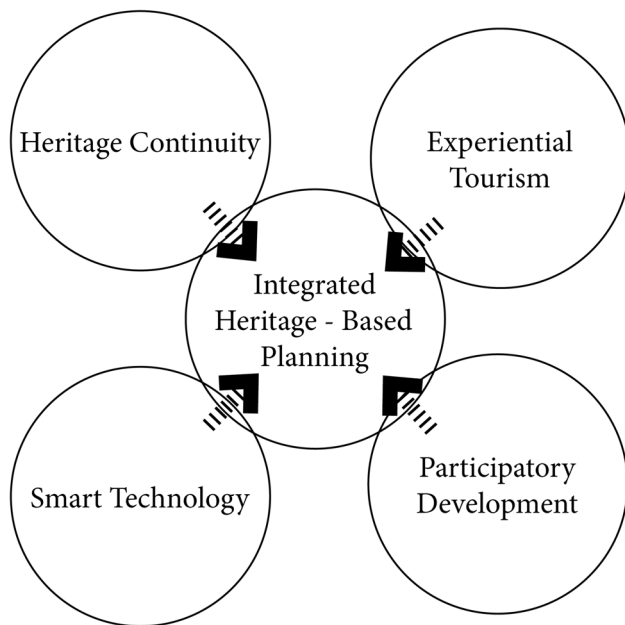


Fig. 9 Cultural landscape preservation framework (HLE: Heritage-Landscape-Ecology)

ecology are managed as a single operational system. Each of these areas contributes to a cohesive framework for sustainable cultural revitalization. Crucially, the framework operationalizes empirical patterns observed in the spatial analysis by retaining in situ production, maintaining short walking links to wharves, and protecting riparian microclimates that enable forming, drying, and firing. Table 3 links each component to its specific contributions to preserving and holistically developing the village's cultural and spatial fabric, supporting sustainable cultural revitalization.

The first element of this model involves heritage continuity. This refers to the ongoing presence of both tangible and intangible cultural assets within the everyday life of the community. In Mang Thit, these assets include the iconic red kilns, riverside drying yards, clay processing methods, and intergenerational knowledge of ceramic

Table 3 Integrated planning matrix linking six solution groups to four components (heritage continuity, experiential tourism, participatory development, smart technology)

6 Solution groups	Heritage continuity	Experiential tourism	Participatory development	Use of smart technology
(1) Map and assess conservation status	● GIS heritage layers, condition criteria	○ Content for guided tours	● Participatory mapping	● GIS database
(2) Designate protected heritage zones	● Core and buffer zoning	○ Route design within allowed areas	● Co-governance rules	● Digital zoning map and compliance alerts
(3) Experiential tourism: workshops and homestays	○ Interpret craft practices	● Itineraries and experience modules	● Co-design products with artisans	● Booking, AR/QR interpretation
(4) Restructure production toward small, symbolic items	● Preserve formal language, glazes, colors	● Make-buy experience bundles	● Benefit sharing, collective mark	● E-commerce, product traceability
(5) Improve non-motorized network, prioritize waterways	○ Manage visitor loads in heritage areas	● Walk-bike-boat network	● Co-design safe routes	● Smart way-finding, footfall counters
(6) Link adjacent functional zones within ~6 km	● Ecological buffer supporting heritage	● Agro-cultural tour extensions	● Coop and smallholder linkages	● Impact dashboard for inter-zone effects

● primary alignment; ○ supporting role.

production. Preserving these features requires managing the living system of spaces, practices, and ecologies, not merely maintaining structures unchanged. According to Guo et al. [19], effective village planning includes conservation zones where traditional spaces remain in use for their original or adapted purposes [19]. Aranburu et al. [3] further highlight that protecting spatial character while allowing for community activity is key to keeping heritage relevant [3]. Consistent with the findings, heritage continuity in Mang Thit is anchored in the repeating kiln-centred clusters and their spatial logic along the river; accordingly, conservation must secure both the fabric and the functioning of these ensembles.

The second part of the framework is experiential tourism. Rather than building isolated museums or exhibition spaces, planning should focus on developing tourism as a cultural journey. Visitors to Mang Thit could take part in pottery workshops, tour operational kilns, listen to local stories, and experience the relationship between ceramic work and

the river landscape. Research by Mai et al. [33] shows that this form of engagement creates deeper cultural understanding and also provides direct income for local families [33]. Tourism becomes most effective when it is rooted in real life and helps extend the craft tradition, not replace it. Accordingly, experiential routes should follow the existing walk-yard-kiln-wharf sequences identified in the spatial analysis, ensuring that visitor flows reinforce rather than fragment everyday production.

A third pillar of planning involves participatory development, ensuring that artisans, families, and community members shape how the village changes over time. Heritage is most meaningful when it is lived and interpreted by those who belong to it [22]. Any changes in the built environment should reflect local architectural styles and values, with guidelines encouraging traditional materials such as tile roofs, unplastered brick, and open courtyards. Preserving these elements strengthens not only aesthetic harmony but also the identity and memory embedded in place. In Mang Thit, participation is organized at the cluster level around the kiln-yard-house or workshop-wharf ensemble, where households, artisan groups, and hamlet leaders co-manage space and production–tourism activities. To formalize this, a Cluster Co-Governance Board (CCB) is recommended, comprising representatives of craft households, women’s and youth groups, wharf owners, the hamlet leader, and the district culture office. The CCB would have clear authorities: co-designing operating norms for clusters (drying hours, shared yard use, fire safety, visitor limits), issuing community wharf regulations (craft-related water access, visitor routing, maintenance fees), jointly reviewing house and workshop renovations against local design codes, and managing a cluster microfinance fund for small workshop upgrades, tools, shading, and venting. In this way, participatory development in Mang Thit becomes a cluster-based co-governance mechanism in which the community decides, implements, and monitors change through specific indicators, rather than a purely aesthetic exercise.

The final component is the use of smart technology. Tools such as geographic information systems, digital mapping, and mobile apps can help manage heritage areas, guide tourists, and communicate rules and stories. These technologies can make planning more transparent and responsive. In Jeonju Hanok Village in South Korea, for example, digital platforms supported both spatial regulation and public engagement [27]. Studies by Shen and Chou [53], as well as Hua et al. [23], suggest that digital heritage tools also help to organize tourism flow and improve infrastructure decisions [23, 53]. For Mang Thit, three initiatives are specified: (1) visitor-flow management with timed entry, footfall counters, and dynamic rerouting along the walk-yard-kiln-wharf sequence; (2) public engagement via a participatory map

and a simple cluster dashboard; and (3) accessible tourism information through QR or light AR at kilns and yards, plus multilingual audio guides and offline maps. Operationally, decisions are tied to indicators from this study, including kiln activity, household craft income, walkable distance to wharves, and riparian buffer condition, with clear thresholds that trigger actions such as rerouting, visitor caps, or maintenance.

To apply this integrated model effectively, it is also important to establish clear planning guidelines. These should address the height and density of new buildings, the protection of traditional materials, and the layout of communal areas. Cu and Le [10] recommend creating village-specific codes that treat cultural space not only as a design matter but also as a shared social structure [10]. Nguyen [37] emphasizes the value of preserving features like communal docks, riverside work areas, and craft alleys, which represent important forms of soft heritage often overlooked in conventional plans [36]. Based on the results, 30–50 m riparian buffers with native vegetation, water-access easements for craft operations, and visitor carrying capacity thresholds along the river corridor are recommended, so that environmental regulation supports the functioning of kiln clusters rather than displacing them.

In conclusion, Mang Thit Village has the potential to become a national example of heritage-led rural development. Its future success depends on whether planning choices can support cultural continuity, meaningful tourism, inclusive governance, and technological innovation. A framework based on these four components offers a pathway for preserving identity while creating new opportunities. Above all, by grounding planning in the clustered linear ensembles and river linkages documented in this research, Mang Thit can move beyond protecting kiln structures and architecture alone to a sustainable cultural landscape that keeps production, community life, and ecology aligned over time.

6.3 Potential conflicts in integrating tourism with traditional livelihoods, and balancing mechanisms

Integrating experiential tourism with traditional livelihoods can revitalize Mang Thit’s economy and reinforce conservation outcomes, yet it also introduces tensions that must be anticipated in planning and governance. The most recurrent risks relate to how market demand reshapes craft practices, how labor and time are reallocated within households, how benefits are distributed across the community, how visitor flows strain sensitive spaces, and how volatility in tourist demand can destabilize livelihoods. In this subsection, these potential conflicts are discussed in relation to the four components of the framework (heritage continuity, experiential

tourism, participatory development, and smart technology) while remaining anchored to the six solution groups set out in Table 3.

A first tension concerns commodification and authenticity. As tourism grows, commercial pressure can push producers toward fast-selling souvenir lines, narrowing design vocabularies, weakening ritual or utilitarian functions, and eroding the perceived authenticity that underpins both cultural value and visitor experience. Recent studies report that higher commercialization in historic towns is associated with lower tourist preference and intent to visit, while perceived authenticity is a strong antecedent of satisfaction, attachment, and revisit intention [32]. To balance this, design codes and authenticity safeguards should be embedded in zoning and production guidance (heritage continuity; solution groups 1, 2, 4), specifying protected forms, materials, glazes, and core steps in clay processing, while experiential programming should privilege interpretation before retail so that workshop encounters reinforce cultural pedagogy rather than displacing it (experiential tourism; solution groups 3, 4). Progress can be tracked through indicators such as the share of products meeting design-code criteria and the proportion of workshops that demonstrably teach core techniques.

A second set of conflicts arises from time and labor diversion as households reallocate effort from production and apprenticeship to visitor-facing tasks. Empowerment scholarship shows how tourism can differentially burden groups, often women, and why role clarity, participation ladders, and power redistribution are essential for equitable outcomes [51]. A participatory development approach can mitigate these effects by establishing compensated and rotating demonstration roles, formalizing apprentice stipends tied to verified production hours, and supporting women's groups for hospitality enterprises (solution groups 3, 4 for role design; 1, 4 for skills transmission). Monitoring weekly hours between production and tourism tasks, apprentice numbers, and women's revenue shares can keep these trade-offs visible and correctable.

Distributional equity presents a third challenge. Without transparent rules, external operators or local elites may capture a disproportionate share of tourism value, while households near prime routes shoulder crowding, noise, and maintenance burdens. Community-based tourism literature documents governance gaps and elite-capture risks and prescribes co-governance and revenue-sharing rules as countermeasures [41]. The framework addresses this through collective marks, minimum price guidelines for demonstrations, cooperative marketing, and village-level grievance channels (participatory development and smart technology; solution group 4), accompanied by benefit-incidence metrics reported publicly.

Spatial pressure is a fourth concern. Visitor flows can overload boat landings, narrow alleys, drying yards, and ritual spaces, displacing production and community practices. Heritage governance is typically characterized by both symbiosis and tension, requiring negotiated trade-offs across space and time [55]. In response, core and buffer zoning with timed access (heritage continuity and experiential tourism; solution groups 2, 5) and network design with smart wayfinding (smart technology; solution group 5) can distribute crowds away from bottlenecks and sensitive nodes. Planners should track peak footfall counts, dwell times in production zones, and logged conflicts, adjusting routes and schedules around kiln cycles and ritual calendars.

Tourism's seasonality and shock sensitivity introduce a fifth risk: cultural distortion and livelihood volatility. Programming to tourist calendars can desynchronize traditional work rhythms, and over-reliance on visitor spending can undermine sustainability when demand falls. Systematic reviews of rural tourism caution against single-pillar revitalization strategies and advocate diversified income bases [63]. Product-line diversification toward non-tourism orders, such as landscape vessels and restoration pieces, and agro-cultural linkages within the six-kilometer buffer (solution groups 4, 6) can smooth income while retaining work rhythms rooted in the craft, guided by resilience indicators such as revenue concentration and month-to-month income variance.

Finally, information asymmetry can undermine all of the above. When planners and communities cannot see who benefits, where pressure concentrates, or when thresholds are crossed, corrective action arrives late. Smart tourism technologies, including heritage GIS layers, AR or QR interpretation, footfall counters, and public dashboards, can support more adaptive management and shift visitor learning away from the most sensitive spaces [46]. In this study's framework, these tools (solution groups 1, 3, 5, 6) are coupled with data-sharing covenants so that anonymized flows and revenue splits inform local decision-making rather than remaining proprietary.

To operationalize these ideas, include a Conflict-Mitigation Matrix (Table 4) that aligns each conflict domain with measurable indicators, planning tools, responsible bodies, data sources, and the corresponding solution groups and framework components. For example, "crowding at boat landings" can be paired with "peak footfall counts" as an indicator, "timed entries and alternate routes" as tools, the "village committee and tourism office" as responsible bodies, and "sensor counts and ticket logs" as data sources, mapped to solution groups 2 and 5 and to the heritage continuity and smart technology components. In this way, conflict management is not an add-on but a design constraint

Table 4 Conflict–Mitigation matrix for craft-village tourism integration

Conflict domain	Monitoring indicators	Planning/governance tools	Responsible bodies	Data sources	Solution groups (1–6)	Framework components (4)
Commodification eroding authenticity	% of products meeting design-code criteria; visitor authenticity ratings	Design code for forms, glazes, colors, core clay-processing; interpretation-before-retail sequencing	Craft-village board; Dept. of Culture; Artisans' guild	Peer review; visitor surveys; random product checks	1, 2, 4	Heritage continuity; Experiential tourism
Time/labor diversion from production and apprenticeship	Weekly hours production vs. tourist tasks; number of apprentices; women's revenue share	Rotated, compensated demo slots; apprentice stipends tied to verified hours; women's groups for hospitality	Commune People's Committee; Cooperative; Women's Union; Artisans' guild	Timesheets; coop ledgers; household surveys	3, 4 (roles); 1, 4 (skills)	Participatory development; Heritage continuity
Inequitable benefit distribution/elite capture	Local value-capture rate (%); number of beneficiary households; grievance resolution time	Revenue-sharing rules; minimum price schedule; collective mark; grievance channel	Cooperative; Community-based tourism team; District Econ. Office	Sales records; service contracts; grievance log	4	Participatory development; Smart technology
Crowding and spatial conflicts at landings/drying yards/sacred spaces	Peak footfall counts; dwell time in production areas; conflicts per month	Core-buffer zoning; timed access aligned with kiln cycles/ritual calendars; walk-bike-boat network	Heritage mgmt. unit; Urban Mgmt. Office; Commune police	Footfall counters; kiln logs; festival calendars	2, 5	Heritage continuity; Experiential tourism; Smart technology
Seasonality and over-reliance on tourism	Revenue concentration (tourism vs. non-tourism); month-to-month income variance	Product-line diversification (restoration/landscape orders); agro-cultural linkages within ~6 km	Cooperative; Agri & Rural Dev. Office; Partner firms	Order books; supply contracts; commune stats	4, 6	Heritage continuity; Experiential tourism
Use-conflicts between production and visiting hours	% of production time interrupted; artisan feedback score	Alternate routes; priority signage for production zones; quiet-time windows	Route committee; Neighborhood self-management group	Workshop logs; QR feedback	5	Experiential tourism; Smart technology
Information asymmetry/low transparency	Dashboard availability; % operators sharing data; policy actions triggered	Smart governance stack: heritage GIS, AR/QR, counters, dashboard; data-sharing covenants	District/Commune gov.; Culture Dept.; IT working group	GIS; booking APIs; counters; surveys	1, 3, 5, 6	Smart technology; Participatory development
Design drift in new buildings	Permits out of code; % traditional materials in new projects	Design code; community design review; priority materials list	Urban Mgmt. Office; Craft guild; Community design panel	Permit files; review minutes	2, 4	Heritage continuity; Participatory development

embedded in the existing matrix, improving both implementability and evaluation across the integrated planning framework.

7 Conclusions

Mang Thit Pottery Village is not only a center of traditional ceramic production but also a living cultural landscape that deeply reflects the identity of the riverine Mekong Delta. In the context of rapid urbanization, industrial competition, and shifting social demands, this heritage faces a serious risk of decline if it is not addressed through systematic and sustainable interventions.

Building on this recognition, the study proposes a system of six integrated solution groups aimed at preserving and comprehensively developing the cultural and spatial fabric of Mang Thit. These solutions include: (1) mapping

and assessing levels of conservation for both tangible and intangible heritage, (2) designating protected heritage zones to safeguard cultural landscapes, (3) developing experiential tourism models such as craft workshops and homestays, (4) restructuring production toward smaller-scale items with symbolic and cultural value for decoration and souvenirs, (5) improving non-motorized transport systems within the village, with emphasis on waterway hubs and internal connections, and (6) expanding links with nearby functional zones, such as vegetable gardens and fruit orchards within a six-kilometer radius, to diversify visitor activities and create ecological buffers. Operationally, the framework is anchored in the clustered linear ensembles identified in the spatial analysis (kiln-drying yard-house/workshop-wharf) and their dependence on riparian conditions, so that conservation secures both fabric and function rather than structures alone.

When implemented through collaboration among government agencies, experts, and residents, Mang Thit can shift from a declining craft settlement to a model of sustainable village development that safeguards heritage while adapting to contemporary demands. In parallel, the study specifies potential conflicts that arise when tourism intersects with traditional livelihoods and proposes practical balancing mechanisms: safeguards against commodification that erodes authenticity; participation and empowerment rules to address time and labor diversion; transparent benefit-sharing to prevent elite capture; core-buffer zoning and timed access to manage crowding in sensitive spaces; diversification strategies to reduce exposure to seasonality; and a smart governance stack for monitoring and adaptive management. See Sect. 4.3 and Table 4 for the Conflict-Mitigation Matrix linking each risk to indicators, tools, responsible bodies, and data sources. By explicitly coupling the six solution groups with conflict-aware governance tools, the framework provides a clear pathway from diagnosis to implementation and evaluation, increasing the likelihood of durable heritage outcomes.

Methodologically, the study relies on qualitative investigation, including spatial interpretation, documentary sources, and cultural analysis. This orientation captures symbolic meanings, the embeddedness of pottery practices in daily life, and community perspectives that are not reducible to quantitative measures. By design, the article does not aim for statistical inference or forecasting; its principal contribution is to identify attributes, variables, and candidate indicators that can inform subsequent quantitative or mixed-methods research, including testable hypotheses and measurement pathways. For planning audiences, these contributions clarify what to measure, where to intervene, and how to monitor trade-offs when tourism and craft production intersect.

Limitations follow from this qualitative design. Findings are interpretive and case-specific rather than standardized measurements, which constrains direct comparability and generalization. Broader quantitative indicators such as production output, demographic trends, or tourism flows were not analyzed here, and long-term projections lie outside the scope. Results should therefore be read as exploratory, with the variables and indicators identified here intended to guide follow-up designs that test, quantify, and generalize observed patterns, for example through household surveys, cooperative records, and ward-level statistics. Because the present study focuses on a single representative case, future work should extend the analysis to other traditional craft villages in the Mekong Delta and across the country. Comparative studies that incorporate the conflict domains and balancing mechanisms in Sect. 4.3 can test, adapt, and scale lessons from Mang Thit, contributing to a

more comprehensive theoretical framework and an inclusive planning ecosystem for heritage-based villages, as well as to broader goals of rural revitalization, cultural sustainability, and the affirmation of regional identity under deeper global integration.

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Declarations

Conflict of interest The author has no conflicts of interest to declare that are relevant to the content of this article.

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